

Lagrange Duality (Part I)

Dual lower bounds, strong duality, KKT, and sensitivity

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Why duality matters

Duality gives a systematic answer to a basic question:

Certification question

Given a candidate feasible point $\bar{x}$, how can we prove that it is **close to optimal** or even exactly optimal?

- ▶ The primal problem gives us **upper bounds**: if $\bar{x}$ is feasible, then

$$p^* \leq f(\bar{x}).$$

- ▶ Duality gives us **lower bounds**: for every dual-feasible pair (λ, μ) ,

$$g(\lambda, \mu) \leq p^*.$$

- ▶ If we ever get

$$f(\bar{x}) = g(\lambda, \mu),$$

then the upper and lower bounds meet, so $\bar{x}$ must be optimal.

왜 Duality 중요한가?

WHY?

최적화 문제를 풀 때는 보통 두 가지를 동시에 알고 싶습니다.

“좋은 해를 찾았는가?” “그 해가 정말 최적인가?”

프라이멀 feasible point는 “이 정도 값은 가능하다”는 **상한**을 주고, 듀얼 feasible point는 “아무리 좋아도 이 값보다 낮을 수 없다”는 **하한**을 줍니다.

SO?

듀얼리티의 목적은 단순히 또 다른 문제를 만드는 것이 아니라, 계산한 해가 최적임을 보장하도록 하는 것

POINT

앞으로 모든 정의는 다음 질문에 답하기 위해 등장합니다:

어떻게 하면 좋은 lower bound를 체계적으로 만들 수 있을까?

Companion notebook and chapter map

Companion notebook

Open in Jupyter: [duality gap notebook](#)

Includes:

- ▶ Lagrangian and dual-function computations
- ▶ duality-gap visualizations
- ▶ numerical examples of strong duality

Roadmap

- 1 Lagrangian and dual function
- 2 Weak vs. strong duality
- 3 Geometric interpretation
- 4 KKT conditions
- 5 Slater's theorem
- 6 Sensitivity / shadow prices

Motivation: optimality gap and the need for lower bounds

Suppose $\bar{x}$ is a feasible point for

$$\min_{x \in \Omega} f(x), \quad p^* = \inf_{x \in \Omega} f(x).$$

Then automatically $p^* \leq f(\bar{x})$.

Optimality gap

The error of $\bar{x}$ is $f(\bar{x}) - p^* \geq 0$.

To control this quantity, we need a **lower bound** on p^* .

Central question

How do we construct a lower bound on p^* that is

- ▶ valid for every feasible point,
- ▶ easy to compute and, ideally, as tight as possible?

The primal problem

Consider the general optimization problem

$$\min f(x) \quad \text{s.t.} \quad f_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_j(x) = 0, \quad j = 1, \dots, p.$$

Feasible set

$$\Omega = \{x \in \mathcal{D} : f_i(x) \leq 0 \quad \forall i, \quad h_j(x) = 0 \quad \forall j\}.$$

Let p^* denote the optimal value. There are three possibilities:

$$p^* = \begin{cases} +\infty, & \text{if the primal is infeasible,} \\ \text{finite,} & \text{if the primal has a finite optimum,} \\ -\infty, & \text{if the primal is unbounded below.} \end{cases}$$

Important point

Duality theory makes sense even when the primal is *not* convex.

The Lagrangian

Definition

For $x \in \mathcal{D}$, $\lambda \in \mathbb{R}^m$, and $\mu \in \mathbb{R}^p$, define

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \mu_j h_j(x).$$

- ▶ λ_i is the multiplier for inequality constraint $f_i(x) \leq 0$.
- ▶ μ_j is the multiplier for equality constraint $h_j(x) = 0$.
- ▶ Intuitively, each multiplier assigns a **price** to violating a constraint.

Interpretation

If $f_i(x) > 0$, then the inequality is violated. A large positive λ_i penalizes that violation inside the Lagrangian.

But why require $\lambda_i \geq 0$ later?

Because if $\lambda_i < 0$, then violating the constraint could *decrease* the Lagrangian, which would destroy the lower-bound property.

How the Lagrangian relaxes constraints

The primal problem is

$$\min_x f(x) \quad \text{s.t.} \quad f_i(x) \leq 0, \quad h_j(x) = 0.$$

The Lagrangian rewrites this as a family of unconstrained problems:

$$x \mapsto \mathcal{L}(x, \lambda, \mu).$$

If x is feasible

$$f_i(x) \leq 0, \quad h_j(x) = 0.$$

So for $\lambda \geq 0$,

$$\mathcal{L}(x, \lambda, \mu) \leq f(x).$$

Thus feasible points do not get penalized upward.

If x is infeasible

At least one $f_i(x) > 0$ or $h_j(x) \neq 0$.

Then $\mathcal{L}(x, \lambda, \mu)$ changes according to the penalties:

$$\lambda_i f_i(x), \quad \mu_j h_j(x).$$

We can use these penalties to create lower bounds on the primal value.

Big picture

The primal problem keeps constraints *hard*. The Lagrangian converts them into *prices*.

The Lagrange dual function

Definition

$$g(\lambda, \mu) = \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \mu) = \inf_{x \in \mathcal{D}} \left(f(x) + \sum_i \lambda_i f_i(x) + \sum_j \mu_j h_j(x) \right).$$

So the dual function is obtained by:

- ❶ fixing the multipliers (λ, μ) ,
 - ❷ minimizing the Lagrangian over x .
- ▶ The result depends only on (λ, μ) .
 - ▶ Each dual-feasible pair gives one candidate lower bound on p^* .

What kind of object is g ?

Even if the primal is nonconvex, the dual function is always **concave**.

Why the dual function is always concave

Fix $x \in \mathcal{D}$. As a function of (λ, μ) ,

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i f_i(x) + \sum_j \mu_j h_j(x)$$

is **affine**.

Key structural fact

$$g(\lambda, \mu) = \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \mu)$$

is the pointwise infimum of a family of affine functions.

We now verify concavity directly.

Take two points $(\lambda^{(1)}, \mu^{(1)})$ and $(\lambda^{(2)}, \mu^{(2)})$, and $\theta \in [0, 1]$. Then

$$\begin{aligned} g(\theta z_1 + (1 - \theta) z_2) &= \inf_x \mathcal{L}(x, \theta z_1 + (1 - \theta) z_2) \\ &= \inf_x \left(\theta \mathcal{L}(x, z_1) + (1 - \theta) \mathcal{L}(x, z_2) \right), \end{aligned}$$

where $z_k = (\lambda^{(k)}, \mu^{(k)})$.

Concavity proof, continued

For every fixed x ,

$$\theta \mathcal{L}(x, z_1) + (1 - \theta) \mathcal{L}(x, z_2) \geq \theta \inf_y \mathcal{L}(y, z_1) + (1 - \theta) \inf_y \mathcal{L}(y, z_2).$$

Now take the infimum over x on the left-hand side:

$$g(\theta z_1 + (1 - \theta) z_2) \geq \theta g(z_1) + (1 - \theta) g(z_2).$$

Conclusion

$$g(\theta z_1 + (1 - \theta) z_2) \geq \theta g(z_1) + (1 - \theta) g(z_2),$$

so g is **concave**.

Important consequence

The dual problem will be a convex optimization problem even when the primal is not.

The lower-bound property: statement

Theorem

For every $\lambda \geq 0$ and every $\mu \in \mathbb{R}^p$,

$$g(\lambda, \mu) \leq p^*.$$

Equivalently, every dual-feasible pair gives a valid lower bound on the primal optimal value.

Why this theorem is fundamental

This one inequality is the entire reason duality is useful:

$$\text{dual feasible pair} \implies \text{provable lower bound on } p^*.$$

We prove it in very small steps.

Lower-bound proof: step 1

Fix any primal-feasible point $x \in \Omega$ and any dual-feasible pair (λ, μ) with $\lambda \geq 0$.

Because x is feasible,

$$f_i(x) \leq 0 \quad \text{for all } i, \quad h_j(x) = 0 \quad \text{for all } j.$$

Multiply the inequalities by nonnegative numbers λ_i :

$$\lambda_i f_i(x) \leq 0 \quad (i = 1, \dots, m).$$

Multiply the equalities by arbitrary μ_j :

$$\mu_j h_j(x) = 0 \quad (j = 1, \dots, p).$$

Add everything to $f(x)$:

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i f_i(x) + \sum_j \mu_j h_j(x) \leq f(x).$$

Meaning

At a feasible point, the Lagrangian cannot exceed the true objective value.

Lower-bound proof: step 2

Now take the infimum over all $x \in \mathcal{D}$:

$$g(\lambda, \mu) = \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \mu).$$

Since $\Omega \subseteq \mathcal{D}$, taking the infimum over the larger set can only decrease the value:

$$g(\lambda, \mu) \leq \inf_{x \in \Omega} \mathcal{L}(x, \lambda, \mu).$$

From the previous slide, for every feasible x ,

$$\mathcal{L}(x, \lambda, \mu) \leq f(x).$$

Therefore

$$\inf_{x \in \Omega} \mathcal{L}(x, \lambda, \mu) \leq \inf_{x \in \Omega} f(x) = p^*.$$

Combining the inequalities,

$$g(\lambda, \mu) \leq p^*.$$

Interpretation

The dual function is a machine that converts multipliers into guaranteed lower bounds.

The Lagrange dual problem

If each (λ, μ) gives a lower bound, the best one is obtained by maximizing over all dual-feasible pairs.

Dual problem

$$\max_{\lambda, \mu} g(\lambda, \mu) \quad \text{s.t.} \quad \lambda \geq 0.$$

Its optimal value is denoted by

$$d^* = \sup_{\lambda \geq 0, \mu \in \mathbb{R}^p} g(\lambda, \mu).$$

- ▶ The primal tries to make the objective *small*.
- ▶ The dual tries to make the lower bound as *large* as possible.

Why the dual is always convex

Because g is concave, maximizing g over the convex set $\{\lambda \geq 0\}$ is a convex optimization problem.

A one-dimensional toy example

Consider

$$\min_x \frac{1}{2}x^2 + x \quad \text{s.t.} \quad x \geq 1.$$

Write the constraint as

$$1 - x \leq 0.$$

Then the Lagrangian is

$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^2 + x + \lambda(1 - x), \quad \lambda \geq 0.$$

Primal optimum

The unconstrained minimizer solves

$$x + 1 = 0 \implies x = -1,$$

but this is infeasible. Hence the constrained optimum is at the boundary:

$$x^* = 1, \quad p^* = \frac{1}{2}(1)^2 + 1 = \frac{3}{2}.$$

Goal

We will compute $g(\lambda)$ explicitly and see how the best lower bound exactly matches p^* .

Toy example: computing the dual function

For fixed λ , the Lagrangian is a quadratic in x :

$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^2 + (1 - \lambda)x + \lambda.$$

To minimize over x , differentiate:

$$\frac{\partial \mathcal{L}}{\partial x} = x + (1 - \lambda) = 0 \quad \implies \quad x(\lambda) = \lambda - 1.$$

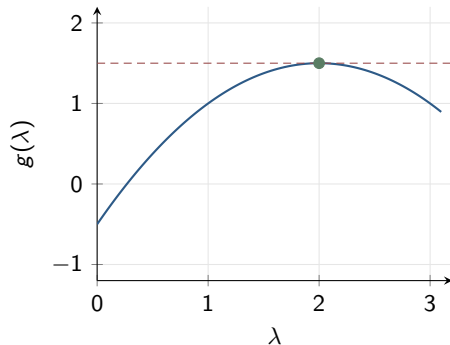
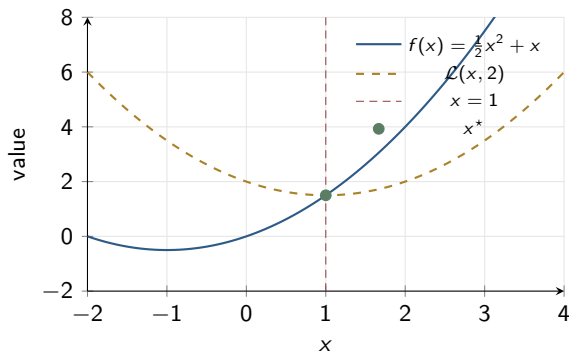
Substitute back:

$$\begin{aligned} g(\lambda) &= \mathcal{L}(\lambda - 1, \lambda) \\ &= \frac{1}{2}(\lambda - 1)^2 + (1 - \lambda)(\lambda - 1) + \lambda \\ &= -\frac{1}{2}\lambda^2 + 2\lambda - \frac{1}{2}. \end{aligned}$$

So the dual problem is

$$\max_{\lambda \geq 0} \left(-\frac{1}{2}\lambda^2 + 2\lambda - \frac{1}{2} \right).$$

Toy example: strong duality in a picture



Conclusion

The dual maximizer is $\lambda^* = 2$, and

$$d^* = g(2) = \frac{3}{2} = p^*.$$

So the lower bound is tight: **strong duality holds**.

Example 1: LP duality

Consider the primal LP

$$(P) \quad \min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

Rewrite $x \geq 0$ as

$$-x \leq 0.$$

Introduce:

- ▶ $\lambda \geq 0$ for the inequality $-x \leq 0$,
- ▶ $\mu \in \mathbb{R}^p$ for the equality $Ax = b$.

Then

$$\mathcal{L}(x, \lambda, \mu) = c^\top x + \lambda^\top (-x) + \mu^\top (Ax - b).$$

We now derive the dual carefully.

LP duality: minimizing the Lagrangian over x

Collect the terms involving x :

$$\begin{aligned}\mathcal{L}(x, \lambda, \mu) &= c^\top x - \lambda^\top x + \mu^\top Ax - b^\top \mu \\ &= -b^\top \mu + x^\top (c - \lambda + A^\top \mu).\end{aligned}$$

Now minimize over unconstrained $x \in \mathbb{R}^d$.

Key dichotomy

If

$$c - \lambda + A^\top \mu = 0,$$

then the linear term in x disappears, so

$$g(\lambda, \mu) = -b^\top \mu.$$

If instead

$$c - \lambda + A^\top \mu \neq 0,$$

then the function is a nonzero linear function of x , whose infimum over $\mathbb{R}^d$ is $-\infty$.

LP duality: eliminating the redundant multiplier

So

$$g(\lambda, \mu) = \begin{cases} -b^\top \mu, & \text{if } c - \lambda + A^\top \mu = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence the dual is

$$\max_{\lambda, \mu} -b^\top \mu \quad \text{s.t.} \quad \lambda \geq 0, \quad c - \lambda + A^\top \mu = 0.$$

Use the equality to eliminate λ :

$$\lambda = c + A^\top \mu.$$

Dual feasibility $\lambda \geq 0$ becomes

$$c + A^\top \mu \geq 0.$$

Therefore the dual LP is

$$\boxed{\max_{\mu} -b^\top \mu \quad \text{s.t.} \quad A^\top \mu + c \geq 0.}$$

Message

The standard LP dual is exactly the Lagrange dual of the primal LP.

Example 2: minimum-norm solution of a linear system

Consider

$$\min_x \|x\|_2^2 = x^\top x \quad \text{s.t.} \quad Ax = b.$$

Only equality constraints are present, so the Lagrangian is

$$\mathcal{L}(x, \mu) = x^\top x + \mu^\top (Ax - b).$$

Step 1: minimize over x

Since $\mathcal{L}$ is a convex quadratic in x , we differentiate:

$$\nabla_x \mathcal{L}(x, \mu) = 2x + A^\top \mu.$$

Set this equal to zero:

$$2x + A^\top \mu = 0 \quad \implies \quad x(\mu) = -\frac{1}{2} A^\top \mu.$$

Next we substitute this minimizer back into the Lagrangian.

Minimum-norm example: substitution step by step

Substitute $x = -\frac{1}{2}A^\top \mu$ into

$$\mathcal{L}(x, \mu) = x^\top x + \mu^\top Ax - b^\top \mu.$$

Compute the pieces:

$$x^\top x = \left(-\frac{1}{2}A^\top \mu\right)^\top \left(-\frac{1}{2}A^\top \mu\right) = \frac{1}{4}\mu^\top AA^\top \mu,$$

and

$$\mu^\top Ax = \mu^\top A \left(-\frac{1}{2}A^\top \mu\right) = -\frac{1}{2}\mu^\top AA^\top \mu.$$

Therefore

$$\begin{aligned} g(\mu) &= \frac{1}{4}\mu^\top AA^\top \mu - \frac{1}{2}\mu^\top AA^\top \mu - b^\top \mu \\ &= -\frac{1}{4}\mu^\top AA^\top \mu - b^\top \mu. \end{aligned}$$

So the dual problem is

$$\boxed{\max_{\mu \in \mathbb{R}^p} \left(-\frac{1}{4}\mu^\top AA^\top \mu - b^\top \mu \right) .}$$

Example 3: minimum norm with a general norm

Now replace $\|x\|_2$ with an arbitrary norm $\|x\|$:

$$\min_x \|x\| \quad \text{s.t.} \quad Ax = b.$$

The Lagrangian is

$$\mathcal{L}(x, \mu) = \|x\| + \mu^\top (Ax - b) = (\|x\| + x^\top A^\top \mu) - b^\top \mu.$$

To compute the dual function, we need the following lemma.

Dual norm lemma

For any y ,

$$\inf_x (\|x\| - y^\top x) = \begin{cases} 0, & \text{if } \|y\|_* \leq 1, \\ -\infty, & \text{if } \|y\|_* > 1. \end{cases}$$

We first prove the lemma, then apply it with $y = -A^\top \mu$.

Proof of the dual norm lemma

Recall the definition of the dual norm:

$$\|y\|_* = \sup_{\|x\| \leq 1} y^\top x.$$

Equivalently,

$$y^\top x \leq \|x\| \|y\|_* \quad \text{for all } x, y.$$

Case 1: $\|y\|_* \leq 1$

Then

$$y^\top x \leq \|x\|,$$

so

$$\|x\| - y^\top x \geq 0 \quad \text{for all } x.$$

Choosing $x = 0$ gives value 0, so the infimum is exactly 0.

Proof of the dual norm lemma (continued)

Case 2: $\|y\|_* > 1$

By definition of the dual norm, there exists some direction v with

$$\|v\| = 1, \quad y^\top v > 1.$$

Set

$$x = tv, \quad t > 0.$$

Then

$$\|x\| - y^\top x = t - t y^\top v = t(1 - y^\top v).$$

Because $y^\top v > 1$, the coefficient $1 - y^\top v$ is negative. So as $t \rightarrow \infty$,

$$t(1 - y^\top v) \rightarrow -\infty.$$

Therefore

$$\inf_x (\|x\| - y^\top x) = -\infty.$$

Conclusion

$$\inf_x (\|x\| - y^\top x) = \begin{cases} 0, & \|y\|_* \leq 1, \\ -\infty, & \|y\|_* > 1. \end{cases}$$

General norm example: dual problem

Apply the lemma with

$$y = -A^\top \mu.$$

Then

$$g(\mu) = \inf_x (\|x\| + x^\top A^\top \mu) - b^\top \mu = \begin{cases} -b^\top \mu, & \text{if } \|A^\top \mu\|_* \leq 1, \\ -\infty, & \text{otherwise.} \end{cases}$$

Hence the dual is

$$\boxed{\max_{\mu} -b^\top \mu \quad \text{s.t.} \quad \|A^\top \mu\|_* \leq 1.}$$

Interpretation

The geometry of the primal norm reappears in the dual as the *dual norm*.

Example 4: duality via conjugate functions

Consider

$$\min_x f(x) \quad \text{s.t.} \quad Ax \leq b, \quad Cx = d.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \lambda^\top (Ax - b) + \mu^\top (Cx - d).$$

Rearrange:

$$\mathcal{L}(x, \lambda, \mu) = f(x) + x^\top (A^\top \lambda + C^\top \mu) - b^\top \lambda - d^\top \mu.$$

So

$$g(\lambda, \mu) = \inf_x \left(f(x) + x^\top (A^\top \lambda + C^\top \mu) \right) - b^\top \lambda - d^\top \mu.$$

Now recall the conjugate:

$$f^*(y) = \sup_x \{x^\top y - f(x)\}.$$

Conjugate example: deriving the dual

Use the identity

$$\inf_x (f(x) + x^\top s) = - \sup_x (x^\top (-s) - f(x)) = -f^*(-s).$$

Here

$$s = A^\top \lambda + C^\top \mu.$$

Therefore

$$g(\lambda, \mu) = -f^*(-A^\top \lambda - C^\top \mu) - b^\top \lambda - d^\top \mu.$$

The dual problem is

$$\boxed{\max_{\lambda, \mu} -f^*(-A^\top \lambda - C^\top \mu) - b^\top \lambda - d^\top \mu \quad \text{s.t.} \quad \lambda \geq 0.}$$

Why this is useful

The primal constraints may be complicated, but the dual feasible set can become very simple: only $\lambda \geq 0$, with the complexity moved into the conjugate term in the objective.

Example 5: a nonconvex primal with a duality gap

Consider the nonconvex problem

$$(P) \quad \min x^4 - 4x^2 \quad \text{s.t.} \quad -1 \leq x \leq 1.$$

Primal value

The unconstrained critical points satisfy

$$4x^3 - 8x = 4x(x^2 - 2) = 0,$$

so the unconstrained minima are at $x = \pm\sqrt{2}$, where the value is -4 .

But these points are infeasible. Over the feasible interval $[-1, 1]$, the minimum occurs at

$$x = \pm 1, \quad f(\pm 1) = 1 - 4 = -3.$$

Hence

$$p^* = -3.$$

We now compute the dual value and see that it is strictly smaller.

Nonconvex example: dual function and duality gap

Introduce multipliers $\mu_1, \mu_2 \geq 0$ for

$$x - 1 \leq 0, \quad -x - 1 \leq 0.$$

Then

$$\mathcal{L}(x, \mu_1, \mu_2) = x^4 - 4x^2 + \mu_1(x - 1) + \mu_2(-x - 1).$$

Let

$$c = \mu_1 - \mu_2.$$

Then

$$g(\mu_1, \mu_2) = \inf_x (x^4 - 4x^2 + cx) - (\mu_1 + \mu_2).$$

At $\mu_1 = \mu_2 = 0$ we have

$$g(0, 0) = \inf_x (x^4 - 4x^2) = -4.$$

For any other feasible pair, the extra term $-(\mu_1 + \mu_2) \leq 0$ can only make the value smaller, so

$$d^* = -4.$$

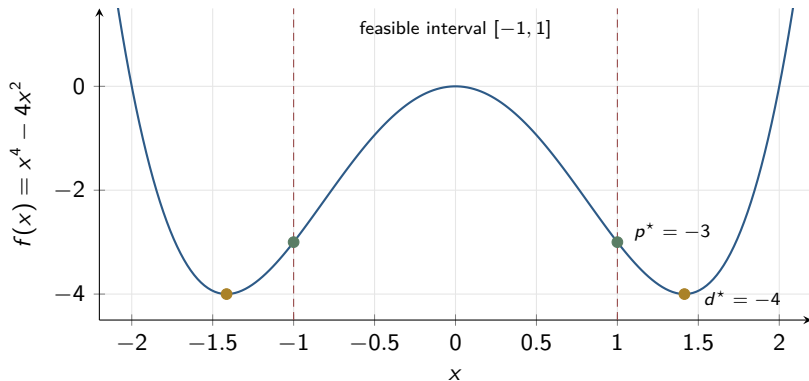
Therefore

$$p^* - d^* = -3 - (-4) = 1 > 0.$$

Takeaway

Duality still works for nonconvex problems, but the dual lower bound may fail to be tight.

The nonconvex gap in a picture



What happened?

The dual function “sees” the unconstrained minima at $\pm\sqrt{2}$, even though the feasible set excludes them. The Lagrange multipliers are not strong enough to close the gap.

Weak duality: the basic inequality

Theorem (Weak duality)

For every primal-feasible x and every dual-feasible (λ, μ) ,

$$g(\lambda, \mu) \leq f(x).$$

Taking the infimum over feasible x and the supremum over dual-feasible (λ, μ) gives

$$d^* \leq p^*.$$

Interpretation

The dual optimum can never exceed the primal optimum. So the duality gap

$$p^* - d^*$$

is always nonnegative.

Weak duality proof: direct inequality proof

Fix a primal-feasible point x and a dual-feasible pair (λ, μ) .

Because x is feasible,

$$f_i(x) \leq 0, \quad h_j(x) = 0.$$

Because (λ, μ) is dual-feasible,

$$\lambda_i \geq 0.$$

Therefore

$$\lambda_i f_i(x) \leq 0, \quad \mu_j h_j(x) = 0.$$

Add these to $f(x)$:

$$\mathcal{L}(x, \lambda, \mu) \leq f(x).$$

Now, by definition of the dual function,

$$g(\lambda, \mu) = \inf_{y \in \mathcal{D}} \mathcal{L}(y, \lambda, \mu) \leq \mathcal{L}(x, \lambda, \mu).$$

Combining the two inequalities,

$$g(\lambda, \mu) \leq \mathcal{L}(x, \lambda, \mu) \leq f(x).$$

That is exactly weak duality.

$$g(\lambda, \mu) \leq f(x) \quad \forall \text{ primal-feasible } x, \quad \forall \text{ dual-feasible } (\lambda, \mu).$$

Weak duality proof: the min-max viewpoint

There is another extremely useful way to read weak duality.

For a fixed x ,

$$\max_{\lambda \geq 0, \mu} \mathcal{L}(x, \lambda, \mu) = \begin{cases} f(x), & \text{if } x \text{ is feasible,} \\ +\infty, & \text{if } x \text{ is infeasible.} \end{cases}$$

Why?

- ▶ If x is feasible, then all penalty terms vanish or are nonpositive, and the best choice is to keep them from increasing the value.
- ▶ If x violates some inequality, send the corresponding $\lambda_i \rightarrow \infty$.
- ▶ If x violates some equality, send the corresponding μ_j to $\pm\infty$ with the correct sign.

So the primal value can be written as

$$p^* = \inf_x \sup_{\lambda \geq 0, \mu} \mathcal{L}(x, \lambda, \mu).$$

The dual value is

$$d^* = \sup_{\lambda \geq 0, \mu} \inf_x \mathcal{L}(x, \lambda, \mu).$$

And the general min-max inequality says

$$\inf_x \sup_z \mathcal{L}(x, z) \geq \sup_z \inf_x \mathcal{L}(x, z),$$

which is exactly

$$p^* \geq d^*.$$

Consequences of weak duality

Immediate corollaries

- 1 If the primal is unbounded below ($p^* = -\infty$), then the dual cannot have a finite feasible point above $-\infty$; in particular $d^* = -\infty$.
- 2 If the dual is unbounded above ($d^* = +\infty$), then the primal must be infeasible ($p^* = +\infty$).
- 3 Every dual-feasible point gives a computable lower bound:

$$g(\lambda, \mu) \leq p^*.$$

Optimization certificate

If you find a primal-feasible x and a dual-feasible (λ, μ) with

$$f(x) = g(\lambda, \mu),$$

then weak duality forces

$$f(x) = p^* = d^* = g(\lambda, \mu).$$

So you have solved both problems at once.

Primal/dual status table under weak duality

	Primal infeasible	Primal finite	Primal unbounded
Dual infeasible	possible	possible	possible
Dual finite	possible	possible	impossible
Dual unbounded	possible	impossible	impossible

Why the red cells are impossible

- ▶ If the primal is unbounded ($p^* = -\infty$), weak duality forbids any finite or $+\infty$ dual optimum.
- ▶ If the dual is unbounded above ($d^* = +\infty$), weak duality forbids any finite or $-\infty$ primal optimum.

Examples of status combinations

Both primal and dual infeasible

$$\min x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 = 1, \quad 2x_1 + 2x_2 = 3.$$

The equalities are contradictory.
Its LP dual also ends up with contradictory equalities, so both sides are infeasible.

Primal finite, dual infeasible

$$\min -x^2 \quad \text{s.t.} \quad -1 \leq x \leq 1.$$

The primal has value $p^* = -1$.
But the Lagrangian is a concave quadratic in x , so its infimum is always $-\infty$. Hence the dual is infeasible.

Moral

Weak duality is universal, but without convexity or constraint qualifications, the dual can behave pathologically.

Another rare case: primal infeasible, dual finite

Consider

$$\min e^x \quad \text{s.t.} \quad e^x = 0.$$

Primal status

This problem is infeasible because $e^x > 0$ for all $x \in \mathbb{R}$. So

$$p^* = +\infty.$$

The Lagrangian is

$$\mathcal{L}(x, \mu) = e^x + \mu e^x = (1 + \mu)e^x.$$

Now minimize over x :

$$g(\mu) = \inf_x (1 + \mu)e^x = \begin{cases} 0, & \mu \geq -1, \\ -\infty, & \mu < -1. \end{cases}$$

Therefore the dual problem is

$$\max_{\mu \geq -1} 0,$$

so

$$d^* = 0.$$

Lesson

It is possible for the primal to be infeasible while the dual still has a finite optimum.

Strong duality

Definition

If

$$p^* = d^*$$

(and both are finite), then we say that **strong duality** holds.

- ▶ Under strong duality, the best lower bound produced by the dual is *exact*.
- ▶ This is the ideal situation: the dual completely certifies the primal optimum.

In linear programming

Strong duality always holds for LPs.

But for general optimization problems, strong duality may fail.

So the next question is:

Under what assumptions do we get $p^* = d^*$?

Why strong duality can fail

Strong duality can fail for several different reasons.

- 1 **Nonconvexity:** the dual lower bound may not be able to capture the true geometry of the primal problem.
- 2 **Degenerate feasible sets:** the feasible set may have empty interior or collapse to a lower-dimensional object.
- 3 **Constraint qualification failure:** even in convex problems, badly behaved constraints can prevent the dual from becoming tight.

Three examples we will discuss

- ▶ an SDP with a positive duality gap,
- ▶ a discrete-domain example,
- ▶ a convex problem whose constraints violate Slater's condition.

Failure example 1: a semidefinite program with a gap

Consider

$$\min y_1 \quad \text{s.t.} \quad \begin{pmatrix} 0 & y_1 & 0 \\ y_1 & y_2 & 0 \\ 0 & 0 & y_1 + 1 \end{pmatrix} \succeq 0.$$

Why must $y_1 = 0$?

For a PSD matrix, every diagonal entry must be nonnegative, and if a diagonal entry is 0, then the corresponding row/column cannot create a negative quadratic form.

Here the upper-left entry is 0. Consider the vector $(1, t, 0)^\top$:

$$(1, t, 0) \begin{pmatrix} 0 & y_1 & 0 \\ y_1 & y_2 & 0 \\ 0 & 0 & y_1 + 1 \end{pmatrix} \begin{pmatrix} 1 \\ t \\ 0 \end{pmatrix} = 2ty_1 + t^2y_2.$$

For this to be nonnegative for every t , we must have $y_1 = 0$.

Then feasibility reduces to $y_2 \geq 0$, so

$$p^* = 0.$$

But the dual value turns out to be $d^* = -1$, giving a gap of 1.

Failure example 2: discrete domain

Let

$$D = [0, 1], \quad f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \in (0, 1]. \end{cases}$$

Consider

$$\min f(x) \quad \text{s.t.} \quad x \leq 0, \quad x \in D.$$

Primal value

The constraints force $x = 0$, so

$$p^* = f(0) = 1.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = f(x) + \lambda x, \quad \lambda \geq 0.$$

Now compute the dual function:

$$g(\lambda) = \inf_{x \in [0, 1]} (f(x) + \lambda x).$$

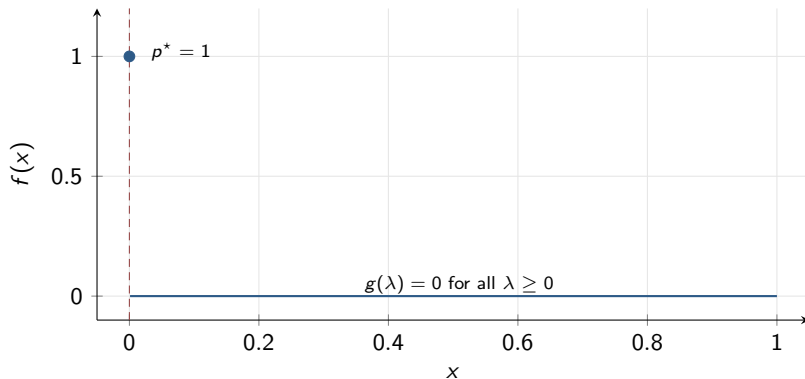
At $x = 0$, the value is 1. For any $x \in (0, 1]$, the value is λx , whose infimum over $(0, 1]$ is 0. Hence

$$g(\lambda) = 0 \quad \text{for every } \lambda \geq 0.$$

Therefore

$$d^* = 0 < 1 = p^*.$$

Discrete-domain gap in a picture



Why the gap appears

The dual relaxation only sees the lower envelope produced by affine penalties. It cannot encode the jump discontinuity that forces the primal to stay at the isolated point $x = 0$.

Failure example 3: constraint qualification failure

Consider

$$\min e^{-x} \quad \text{s.t.} \quad \frac{x^2}{y} \leq 0, \quad y > 0.$$

Primal value

Because $y > 0$, the constraint $\frac{x^2}{y} \leq 0$ forces

$$x^2 \leq 0 \implies x = 0.$$

Any $y > 0$ is then feasible, so

$$p^* = e^0 = 1.$$

The Lagrangian is

$$\mathcal{L}(x, y, \lambda) = e^{-x} + \lambda \frac{x^2}{y}, \quad \lambda \geq 0.$$

Now choose $y = x^3$ and let $x \rightarrow +\infty$. Then

$$e^{-x} + \lambda \frac{x^2}{x^3} = e^{-x} + \frac{\lambda}{x} \rightarrow 0.$$

Hence

$$g(\lambda) = 0 \quad \forall \lambda \geq 0, \quad d^* = 0.$$

So there is a positive duality gap:

$$p^* - d^* = 1.$$

Slater's condition

Now we state the most important sufficient condition for strong duality in convex optimization.

Convex primal problem

$$\min f(x) \quad \text{s.t.} \quad f_i(x) \leq 0, \quad i = 1, \dots, m, \quad Ax = b,$$

where f and each f_i are convex.

Slater's condition

There exists some $\bar{x} \in \text{int}(\mathcal{D})$ such that

$$f_i(\bar{x}) < 0 \quad \forall i, \quad A\bar{x} = b.$$

In words: the inequality constraints can be satisfied **strictly**, while the equalities are satisfied exactly.

Nickname

Slater's condition is often called **strict feasibility**.

Why Slater's condition is a natural assumption

Slater's condition says the feasible set has **room to breathe**.

What it rules out

- ▶ feasible sets consisting only of boundary points,
- ▶ feasible sets with empty interior relative to the inequality constraints,
- ▶ pathological situations where the supporting hyperplane at the optimum becomes vertical.

Why it is often easy to satisfy

If your constraints are ordinary convex inequalities like

$$x > 0, \quad \|x\|_2 < 1, \quad a_i^\top x < b_i,$$

then checking strict feasibility is often straightforward.

But note

Slater's condition is about *inequalities*. Equality constraints are always satisfied exactly, never strictly.

Slater's theorem

Theorem (Slater)

For a convex optimization problem, if

- ▶ the optimal value p^* is finite, and
- ▶ Slater's condition holds,

then

$$p^* = d^*.$$

Moreover, the dual optimum is attained by some (λ^*, μ^*) .

Interpretation

Under strict feasibility, the dual lower bound is not merely valid — it is **exact**.

This is the gateway to KKT

Once strong duality holds, primal and dual optima become tightly linked. The KKT conditions will express that linkage algebraically.

Why Slater implies strong duality: geometric intuition

For convex problems, define the epigraph-style set

$$\mathcal{A} = \{(u, t) : \exists x \in \mathcal{D} \text{ with } f_i(x) \leq u_i, Ax = b, f(x) \leq t\}.$$

- ▶ The primal optimum p^* is the smallest t with $(0, t) \in \mathcal{A}$.
- ▶ The dual problem searches for supporting hyperplanes under $\mathcal{A}$.

What can go wrong without Slater?

The supporting hyperplane at $(0, p^*)$ might be **vertical**. A vertical hyperplane does not correspond to a useful dual multiplier vector.

What Slater prevents

If there is a strictly feasible point, then $\mathcal{A}$ extends into the region $u < 0$. That makes a vertical supporting hyperplane impossible, forcing a nonvertical hyperplane and hence a tight dual certificate.

The set $\mathcal{G}$ and the primal problem

Define

$$\mathcal{G} = \left\{ \left(f_1(x), \dots, f_m(x), h_1(x), \dots, h_p(x), f(x) \right) : x \in \mathcal{D} \right\}.$$

Write a typical point as (u, v, t) , where

$$u \in \mathbb{R}^m, \quad v \in \mathbb{R}^p, \quad t \in \mathbb{R}.$$

Then:

- ▶ u records the inequality-constraint values,
- ▶ v records the equality-constraint values,
- ▶ t records the objective value.

Primal value from the geometry

The primal value is

$$p^* = \inf \{ t : (u, v, t) \in \mathcal{G}, u \leq 0, v = 0 \}.$$

So we look at the slice of $\mathcal{G}$ satisfying the constraints and read off the lowest possible t -coordinate.

The dual function as a supporting hyperplane

For any $\lambda \geq 0$ and $\mu \in \mathbb{R}^p$,

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \lambda^\top u + \mu^\top v = (\lambda, \mu, 1)^\top (u, v, t).$$

So minimizing the Lagrangian over x means

$$g(\lambda, \mu) = \inf_{(u, v, t) \in \mathcal{G}} (\lambda, \mu, 1)^\top (u, v, t).$$

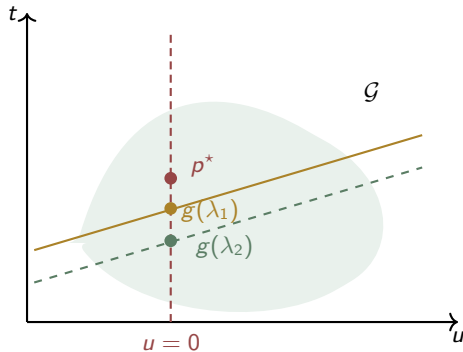
Geometric meaning

For each normal vector $(\lambda, \mu, 1)$, the dual function gives the lowest hyperplane of that slope that still lies below $\mathcal{G}$.

Why this is a lower bound

The hyperplane lies below $\mathcal{G}$, so its intercept at the feasible slice must be below the primal optimum. That is weak duality in geometric language.

A picture: supporting lines and lower bounds



Reading the figure

Each supporting line gives a lower bound. The best one is the highest intercept among all dual-feasible slopes.

The epigraph set $\mathcal{A}$

For convex problems, a more natural set is

$$\mathcal{A} = \{(u, v, t) : \exists x \in \mathcal{D} \text{ such that } f_i(x) \leq u_i, h_j(x) = v_j, f(x) \leq t\}.$$

Why $\mathcal{A}$ is useful

- ▶ $\mathcal{A}$ is an “upper-right” enlargement of $\mathcal{G}$.
- ▶ If the primal is convex, then $\mathcal{A}$ is a **convex set**.
- ▶ The primal value is

$$p^* = \inf\{t : (0, 0, t) \in \mathcal{A}\}.$$

Dual interpretation

The dual function can also be viewed as the supporting hyperplane value of $\mathcal{A}$ with normal vector $(\lambda, \mu, 1)$.

Weak duality and strong duality from the picture

Weak duality

Every supporting hyperplane to $\mathcal{A}$ lies below the set. So its intercept at $(u, v) = (0, 0)$ must satisfy

$$g(\lambda, \mu) \leq p^*.$$

Strong duality

If there exists a **nonvertical** supporting hyperplane to $\mathcal{A}$ at the point $(0, 0, p^*)$, then its intercept equals p^* . That hyperplane corresponds to an optimal dual pair, so

$$d^* = p^*.$$

Role of Slater

Slater's condition forces the supporting hyperplane at $(0, 0, p^*)$ to be nonvertical.

Proof strategy under Slater: one inequality constraint

To keep notation light, we prove strong duality for

$$\min_x f(x) \quad \text{s.t.} \quad f_1(x) \leq 0, \quad x \in \mathcal{D},$$

where f and f_1 are convex and $\mathcal{D}$ is convex.

Define

$$\mathcal{A} = \{(u, t) : \exists x \in \mathcal{D} \text{ with } f_1(x) \leq u, f(x) \leq t\}.$$

Goal

Show that there exists some $\lambda^* \geq 0$ such that

$$g(\lambda^*) = p^*.$$

Roadmap

- 1 Separate $\mathcal{A}$ from the ray below $(0, p^*)$.
- 2 Analyze the separating hyperplane.
- 3 Show its t -coefficient cannot be zero (this is where Slater is used).
- 4 Normalize the hyperplane to obtain a dual multiplier.

Step 1: define the set below the optimum

Let

$$\mathcal{B} = \{(0, s) : s < p^*\}.$$

This is the vertical ray strictly below the point $(0, p^*)$ on the t -axis.

Claim

$$\mathcal{A} \cap \mathcal{B} = \emptyset.$$

Proof.

If $(0, s) \in \mathcal{A}$, then by definition there exists some $x \in \mathcal{D}$ such that

$$f_1(x) \leq 0, \quad f(x) \leq s.$$

So x is primal-feasible, and its objective value is at most s .

If also $s < p^*$, then we have found a feasible point with value strictly below the optimal value — impossible.

Therefore $(0, s)$ with $s < p^*$ cannot belong to $\mathcal{A}$. □

Step 2: apply the separating hyperplane theorem

We now know that:

- ▶ $\mathcal{A}$ is convex,
- ▶ $\mathcal{B}$ is convex,
- ▶ $\mathcal{A}$ and $\mathcal{B}$ are disjoint.

So the separating hyperplane theorem gives numbers $(\lambda, \nu) \neq (0, 0)$ and $a \in \mathbb{R}$ such that

$$\lambda u + \nu t \geq a \quad \forall (u, t) \in \mathcal{A},$$

and

$$\nu s \leq a \quad \forall s < p^*.$$

What this means

The line

$$\lambda u + \nu t = a$$

separates the feasible epigraph set $\mathcal{A}$ from the ray strictly below the optimum.

Step 3: show that $\lambda \geq 0$ and $\nu \geq 0$

Because $\mathcal{A}$ is an epigraph-style set, it is monotone in the northeast direction: if $(u, t) \in \mathcal{A}$, then every (u', t') with

$$u' \geq u, \quad t' \geq t$$

also lies in $\mathcal{A}$.

Now suppose $\lambda < 0$. Then for any fixed $(u, t) \in \mathcal{A}$, we can increase u to $+\infty$, and

$$\lambda u + \nu t \rightarrow -\infty,$$

which would violate the inequality

$$\lambda u + \nu t \geq a.$$

So $\lambda < 0$ is impossible. Hence

$$\lambda \geq 0.$$

Similarly, if $\nu < 0$, increasing t to $+\infty$ would violate separation. So

$$\nu \geq 0.$$

Conclusion

The separating hyperplane has a nonnegative normal:

$$\lambda \geq 0, \quad \nu \geq 0.$$

Step 4: compare the separating line to the primal optimum

From

$$\nu s \leq a \quad \forall s < p^*,$$

take $s \uparrow p^*$ to obtain

$$\nu p^* \leq a.$$

Now take any $x \in \mathcal{D}$. The point

$$(f_1(x), f(x))$$

belongs to $\mathcal{A}$, so separation gives

$$\lambda f_1(x) + \nu f(x) \geq a.$$

Combining with $\nu p^* \leq a$ yields

$$\lambda f_1(x) + \nu f(x) \geq \nu p^* \quad \forall x \in \mathcal{D}.$$

This is the crucial inequality. Everything now depends on whether $\nu > 0$ or $\nu = 0$.

Step 5A: if $\nu > 0$, normalize and get the dual bound

Assume first that $\nu > 0$.

Divide the inequality

$$\lambda f_1(x) + \nu f(x) \geq \nu p^*$$

by ν :

$$f(x) + \frac{\lambda}{\nu} f_1(x) \geq p^* \quad \forall x \in \mathcal{D}.$$

Define

$$\lambda^* = \frac{\lambda}{\nu} \geq 0.$$

Then

$$\mathcal{L}(x, \lambda^*) = f(x) + \lambda^* f_1(x) \geq p^* \quad \forall x \in \mathcal{D}.$$

Take the infimum over x :

$$g(\lambda^*) = \inf_x \mathcal{L}(x, \lambda^*) \geq p^*.$$

But weak duality always gives

$$g(\lambda^*) \leq p^*.$$

Hence

$$g(\lambda^*) = p^*.$$

Done

We have found a dual-feasible point attaining the primal value, so strong duality holds.

Step 5B: if $\nu = 0$, Slater gives a contradiction

Now suppose instead that

$$\nu = 0.$$

Then the key inequality becomes

$$\lambda f_1(x) \geq 0 \quad \forall x \in \mathcal{D}.$$

Since Slater's condition holds, there exists some strictly feasible $\bar{x}$ such that

$$f_1(\bar{x}) < 0.$$

Substitute $x = \bar{x}$:

$$\lambda f_1(\bar{x}) \geq 0.$$

But we already know $\lambda \geq 0$, and $f_1(\bar{x}) < 0$, so the only way

$$\lambda f_1(\bar{x}) \geq 0$$

can hold is if

$$\lambda = 0.$$

Thus

$$(\lambda, \nu) = (0, 0),$$

which contradicts the separating hyperplane theorem.

Conclusion

The case $\nu = 0$ is impossible. Therefore $\nu > 0$, and the previous slide applies.

What the proof really used

Let's summarize the logic.

- ① Separation gives a supporting line at the optimum.
- ② To turn that line into a dual multiplier, we need its t -coefficient ν to be positive.
- ③ Slater's condition forces exactly that by ruling out a vertical supporting line.

Conceptual takeaway

Strong duality is fundamentally a **geometric separation statement**.

Extension to many inequalities

For m inequality constraints, the same proof works in $\mathbb{R}^{m+1}$. The scalar λ is replaced by a vector $\lambda \in \mathbb{R}_{\geq 0}^m$, and the same normalization argument yields an optimal dual vector λ^* .

From strong duality to complementary slackness

Suppose strong duality holds, and let x^* be primal optimal and (λ^*, μ^*) dual optimal. Then

$$f(x^*) = p^* = d^* = g(\lambda^*, \mu^*).$$

Now compare this to the Lagrangian:

$$g(\lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*).$$

Because x^* is feasible,

$$\mathcal{L}(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_i \lambda_i^* f_i(x^*) + \sum_j \mu_j^* h_j(x^*) \leq f(x^*),$$

since $h_j(x^*) = 0$ and $\lambda_i^* f_i(x^*) \leq 0$.

So we have a chain

$$f(x^*) \leq \mathcal{L}(x^*, \lambda^*, \mu^*) \leq f(x^*),$$

hence everything is equal.

Complementary slackness

From the equality

$$\mathcal{L}(x^*, \lambda^*, \mu^*) = f(x^*),$$

we obtain

$$\sum_{i=1}^m \lambda_i^* f_i(x^*) = 0.$$

But every term in the sum is nonpositive:

$$\lambda_i^* \geq 0, \quad f_i(x^*) \leq 0.$$

A sum of nonpositive numbers can equal 0 only if each term is individually zero. Therefore

$$\lambda_i^* f_i(x^*) = 0 \quad \text{for every } i.$$

Interpretation

Either

- ▶ the constraint is **active**: $f_i(x^*) = 0$, in which case λ_i^* may be positive; or
- ▶ the constraint is **inactive**: $f_i(x^*) < 0$, in which case necessarily $\lambda_i^* = 0$.

The KKT conditions

KKT system

A triple (x, λ, μ) satisfies the Karush-Kuhn-Tucker conditions if:

① **Primal feasibility:**

$$f_i(x) \leq 0, \quad h_j(x) = 0.$$

② **Dual feasibility:**

$$\lambda_i \geq 0.$$

③ **Complementary slackness:**

$$\lambda_i f_i(x) = 0.$$

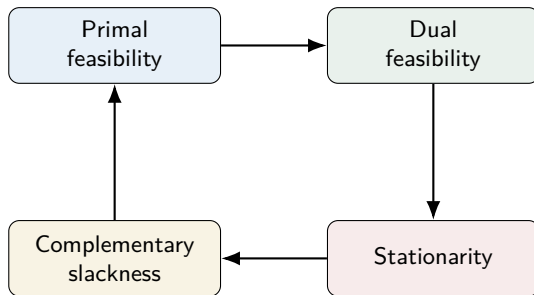
④ **Stationarity:**

$$\nabla f(x) + \sum_i \lambda_i \nabla f_i(x) + \sum_j \mu_j \nabla h_j(x) = 0.$$

Mnemonic

feasible primal + feasible dual + no wasted multiplier + zero gradient.

KKT conditions as a flow diagram



How to read this

At optimum, these four conditions must hold simultaneously. In convex problems with strong duality, this system is equivalent to optimality.

The KKT-optimality theorem

Theorem

For convex optimization problems:

- 1 If strong duality holds and (x^*, λ^*, μ^*) are primal-dual optimal, then they satisfy KKT.
- 2 If a triple $(\bar{x}, \bar{\lambda}, \bar{\mu})$ satisfies KKT, then $\bar{x}$ is primal optimal and $(\bar{\lambda}, \bar{\mu})$ is dual optimal.
- 3 Therefore, under Slater's condition,

$$\bar{x} \text{ is primal optimal} \iff \exists(\bar{\lambda}, \bar{\mu}) \text{ such that KKT holds.}$$

We prove both directions carefully.

Proof that strong duality implies KKT

Suppose x^* and (λ^*, μ^*) are primal and dual optimal, and strong duality holds.

We already showed:

$$f(x^*) = g(\lambda^*, \mu^*) = \inf_x \mathcal{L}(x, \lambda^*, \mu^*) = \mathcal{L}(x^*, \lambda^*, \mu^*).$$

This gives two facts immediately:

- 1 Since x^* attains the infimum of the Lagrangian,

$$x^* \in \arg \min_x \mathcal{L}(x, \lambda^*, \mu^*).$$

Because the problem is differentiable,

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0.$$

That is stationarity.

- 2 Since

$$\mathcal{L}(x^*, \lambda^*, \mu^*) = f(x^*),$$

we get

$$\sum_i \lambda_i^* f_i(x^*) = 0,$$

which yields complementary slackness term by term.

Primal feasibility holds because x^* is primal optimal; dual feasibility holds because (λ^*, μ^*) is dual optimal.

Proof that KKT implies optimality: the main idea

Now suppose $(\bar{x}, \bar{\lambda}, \bar{\mu})$ satisfies KKT.

We want to prove that

$$f(\bar{x}) = p^* = d^* = g(\bar{\lambda}, \bar{\mu}).$$

The proof has two ingredients:

- 1 Stationarity says that $\bar{x}$ minimizes the Lagrangian $\mathcal{L}(x, \bar{\lambda}, \bar{\mu})$.
- 2 Complementary slackness and primal feasibility say that, at $\bar{x}$, the Lagrangian equals the primal objective.

If both statements hold, then the dual function evaluated at $(\bar{\lambda}, \bar{\mu})$ exactly equals the primal value at $\bar{x}$. Weak duality will then force optimality.

KKT implies optimality: step-by-step proof

Because the problem is convex and $\bar{\lambda}_i \geq 0$, the Lagrangian

$$x \mapsto \mathcal{L}(x, \bar{\lambda}, \bar{\mu}) = f(x) + \sum_i \bar{\lambda}_i f_i(x) + \sum_j \bar{\mu}_j h_j(x)$$

is a convex function of x .

By stationarity,

$$\nabla_x \mathcal{L}(\bar{x}, \bar{\lambda}, \bar{\mu}) = 0.$$

For a differentiable convex function, zero gradient means global minimizer. Hence

$$g(\bar{\lambda}, \bar{\mu}) = \inf_x \mathcal{L}(x, \bar{\lambda}, \bar{\mu}) = \mathcal{L}(\bar{x}, \bar{\lambda}, \bar{\mu}).$$

Now expand:

$$\mathcal{L}(\bar{x}, \bar{\lambda}, \bar{\mu}) = f(\bar{x}) + \sum_i \bar{\lambda}_i f_i(\bar{x}) + \sum_j \bar{\mu}_j h_j(\bar{x}).$$

By primal feasibility,

$$h_j(\bar{x}) = 0.$$

By complementary slackness,

$$\bar{\lambda}_i f_i(\bar{x}) = 0.$$

So

$$g(\bar{\lambda}, \bar{\mu}) = f(\bar{x}).$$

KKT implies optimality: finishing the proof

We have just shown

$$g(\bar{\lambda}, \bar{\mu}) = f(\bar{x}).$$

Now compare this to primal and dual optima.

- ▶ Since $\bar{x}$ is primal-feasible,

$$f(\bar{x}) \geq p^*.$$

- ▶ Since $(\bar{\lambda}, \bar{\mu})$ is dual-feasible,

$$g(\bar{\lambda}, \bar{\mu}) \leq d^*.$$

- ▶ By weak duality,

$$d^* \leq p^*.$$

Putting these together:

$$p^* \leq f(\bar{x}) = g(\bar{\lambda}, \bar{\mu}) \leq d^* \leq p^*.$$

Therefore all inequalities are equalities:

$$f(\bar{x}) = p^* = d^* = g(\bar{\lambda}, \bar{\mu}).$$

Conclusion

A KKT point is automatically a primal-dual optimal pair.

Under Slater: KKT $\iff$ optimality

Now combine the last theorem with Slater's theorem.

- ▶ Slater $\implies$ strong duality.
- ▶ Strong duality $\implies$ every primal-dual optimum satisfies KKT.
- ▶ KKT $\implies$ primal-dual optimality.

So for convex problems satisfying Slater's condition:

$$x^* \text{ is optimal } \iff \exists(\lambda^*, \mu^*) \text{ such that KKT holds.}$$

Why this is important computationally

Instead of directly solving the constrained optimization problem, we can often solve the system

$$\begin{cases} f_i(x) \leq 0, \\ h_j(x) = 0, \\ \lambda_i \geq 0, \\ \lambda_i f_i(x) = 0, \\ \nabla f(x) + \sum_i \lambda_i \nabla f_i(x) + \sum_j \mu_j \nabla h_j(x) = 0. \end{cases}$$

Example: unconstrained optimization

For

$$\min_{x \in \mathbb{R}^n} f(x),$$

there are no constraints.

So the Lagrangian is just

$$\mathcal{L}(x) = f(x),$$

and KKT reduces to a single condition:

$$\nabla f(x) = 0.$$

Interpretation

KKT generalizes the familiar first-order condition from calculus.

Convex case

If f is convex, then every solution of

$$\nabla f(x) = 0$$

is globally optimal.

Example: equality-constrained quadratic program

Consider

$$\min_x \frac{1}{2}x^\top Px + q^\top x + r \quad \text{s.t.} \quad Ax = b.$$

The Lagrangian is

$$\mathcal{L}(x, \mu) = \frac{1}{2}x^\top Px + q^\top x + r + \mu^\top (Ax - b).$$

KKT gives:

$$Ax = b, \quad Px + q + A^\top \mu = 0.$$

These can be written as one linear system:

$$\boxed{\begin{pmatrix} P & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x \\ \mu \end{pmatrix} = \begin{pmatrix} -q \\ b \end{pmatrix} .}$$

Why this matters

Many interior-point and primal-dual methods repeatedly solve linear systems of this exact KKT form.

Example: entropy minimization

Consider

$$\min_x \sum_{i=1}^n x_i \log x_i \quad \text{s.t.} \quad \sum_{i=1}^n x_i = 1, \quad Ax \leq b, \quad x_i > 0.$$

Introduce multipliers:

- ▶ $\lambda \geq 0$ for $Ax \leq b$,
- ▶ μ for $\sum_i x_i = 1$.

The Lagrangian is

$$\mathcal{L}(x, \lambda, \mu) = \sum_{i=1}^n x_i \log x_i + \lambda^\top (Ax - b) + \mu \left(\sum_{i=1}^n x_i - 1 \right).$$

We now minimize with respect to each coordinate x_i .

Entropy minimization: stationarity and dual form

Differentiate with respect to x_i :

$$\frac{\partial \mathcal{L}}{\partial x_i} = 1 + \log x_i + \mathbf{a}_i^\top \lambda + \mu,$$

where $\mathbf{a}_i$ is the i -th column of A .

Set the derivative to zero:

$$1 + \log x_i + \mathbf{a}_i^\top \lambda + \mu = 0.$$

Solve for x_i :

$$\log x_i = -(\mathbf{a}_i^\top \lambda + \mu + 1) \implies x_i = \exp(-(\mathbf{a}_i^\top \lambda + \mu + 1)).$$

So once we know the dual solution (λ^*, μ^*) , the primal solution is recovered explicitly:

$$x_i^* = \exp(-(\mathbf{a}_i^\top \lambda^* + \mu^* + 1)).$$

Message

Sometimes the dual is easier to solve, and then the primal optimum drops out from stationarity.

Practical recipe: solve the primal through the dual

When strong duality holds, there are two equivalent routes to the answer.

Route A: solve the primal directly

Find

$$x^* \in \arg \min \{f(x) : f_i(x) \leq 0, h_j(x) = 0\}.$$

Route B: solve the dual, then recover x^*

① Solve

$$\max_{\lambda \geq 0, \mu} g(\lambda, \mu)$$

to obtain (λ^*, μ^*) .

② Minimize the Lagrangian

$$x^* \in \arg \min_x \mathcal{L}(x, \lambda^*, \mu^*).$$

③ Use complementary slackness and feasibility to verify the solution.

Why this can be attractive

The dual may have simpler constraints, lower dimension, or better numerical structure.

Sensitivity interpretation of the dual variables

So far, λ^* was just the multiplier that makes the dual lower bound tight.
But it has a deeper meaning:

it measures how much the optimal value changes when we relax the constraints.

Define the perturbed value function

$$p^*(u) = \inf_x \{f(x) : f_i(x) \leq u_i, i = 1, \dots, m\}.$$

The original problem is recovered at $u = 0$:

$$p^*(0) = p^*.$$

Interpretation of u_i

Increasing u_i relaxes the i -th constraint, so we expect the optimal value to go down (or stay unchanged).

Shadow price theorem

Suppose strong duality holds for the original problem, and let λ^* be an optimal dual multiplier.

Global lower-supporting inequality

For every perturbation u ,

$$p^*(u) \geq p^*(0) - (\lambda^*)^\top u.$$

Local sensitivity

If $p^*(u)$ is differentiable at $u = 0$, then

$$\frac{\partial p^*}{\partial u_i}(0) = -\lambda_i^*.$$

Economic interpretation

λ_i^* is the **shadow price** of the i -th constraint: it is the marginal improvement in the optimal objective per unit relaxation of that constraint.

Proof of the shadow price inequality

Consider the perturbed problem

$$p^*(u) = \inf_x \{f(x) : f_i(x) \leq u_i\}.$$

Its Lagrangian is

$$\mathcal{L}_u(x, \lambda) = f(x) + \sum_i \lambda_i (f_i(x) - u_i).$$

So the perturbed dual function is

$$g_u(\lambda) = \inf_x \mathcal{L}_u(x, \lambda) = \inf_x \left(f(x) + \sum_i \lambda_i f_i(x) \right) - \lambda^\top u = g(\lambda) - \lambda^\top u.$$

By weak duality for the perturbed problem,

$$p^*(u) \geq g_u(\lambda) \quad \forall \lambda \geq 0.$$

Now plug in the optimal multiplier λ^* from the unperturbed problem:

$$p^*(u) \geq g(\lambda^*) - (\lambda^*)^\top u.$$

Strong duality at $u = 0$ gives

$$g(\lambda^*) = p^*(0).$$

Hence

$$\boxed{p^*(u) \geq p^*(0) - (\lambda^*)^\top u.}$$

Why the derivative equals $-\lambda^*$

The inequality

$$p^*(u) \geq p^*(0) - (\lambda^*)^\top u$$

says that the affine function

$$u \mapsto p^*(0) - (\lambda^*)^\top u$$

lies below the perturbation function everywhere and touches it at $u = 0$.

Geometric meaning

This affine function is a **supporting hyperplane** to $p^*(u)$ at the origin.

If p^* is differentiable at $u = 0$, then it has a unique supporting hyperplane there — namely its tangent plane. So the gradient must equal the slope of that affine support:

$$\nabla p^*(0) = -\lambda^*.$$

Coordinate-wise,

$$\frac{\partial p^*}{\partial u_i}(0) = -\lambda_i^*.$$

Interpretation

If λ_i^* is large, then relaxing the i -th constraint decreases the optimal value rapidly.

Shadow-price example: quadratic cost with one inequality

Consider

$$\min x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \leq b.$$

We treat b as the perturbation parameter.

Case 1: $b \geq 0$

The unconstrained minimizer is $(0,0)$, and it is feasible because

$$0 + 0 \leq b.$$

So

$$x^* = (0,0), \quad p^*(b) = 0.$$

The constraint is inactive, so the shadow price should be

$$\lambda^* = 0.$$

Case 2: $b < 0$

Now the unconstrained minimizer is infeasible, so the constraint becomes active.

Shadow-price example: solving the active case

For $b < 0$, form the Lagrangian

$$\mathcal{L}(x_1, x_2, \lambda) = x_1^2 + x_2^2 + \lambda(x_1 + x_2 - b), \quad \lambda \geq 0.$$

Stationarity gives

$$2x_1 + \lambda = 0, \quad 2x_2 + \lambda = 0.$$

So

$$x_1 = x_2 = -\frac{\lambda}{2}.$$

Because the constraint is active,

$$x_1 + x_2 = b.$$

Substitute:

$$-\frac{\lambda}{2} - \frac{\lambda}{2} = b \implies -\lambda = b.$$

Hence

$$\lambda^* = -b > 0.$$

The primal optimizer is

$$x_1^* = x_2^* = \frac{b}{2},$$

and the value is

$$p^*(b) = \left(\frac{b}{2}\right)^2 + \left(\frac{b}{2}\right)^2 = \frac{b^2}{2}.$$

Shadow-price example: checking the derivative

For $b < 0$ we found

$$p^*(b) = \frac{b^2}{2}, \quad \lambda^* = -b.$$

Differentiate:

$$\frac{d}{db} p^*(b) = b.$$

But

$$-\lambda^* = -(-b) = b.$$

So indeed

$$\frac{d}{db} p^*(b) = -\lambda^*.$$

Interpretation

When b increases a little (the constraint relaxes), the optimal cost decreases at rate λ^* .

Economic language

λ^* tells us the value of one extra unit of capacity in the constraint $x_1 + x_2 \leq b$.

보충 설명: shadow price 예제의 active/inactive 해석

$b \geq 0$ 인 경우

원점 $(0, 0)$ 이 feasible이므로 제약 $x_1 + x_2 \leq b$ 는 optimum을 막지 않습니다. 따라서

$$\lambda^* = 0, \quad p^*(b) = 0.$$

제약을 조금 더 완화해도 value가 변하지 않습니다.

$b < 0$ 인 경우

원점이 infeasible이므로 constraint boundary 위에서 최적해가 결정됩니다. 이때

$$\lambda^* = -b > 0, \quad p^*(b) = \frac{b^2}{2}.$$

제약을 완화하면 즉시 objective가 줄어듭니다.

핵심

shadow price는 “현재 optimum에서 실제로 병목이 되는 제약”에만 양수가 됩니다.

Summary of the chapter

- ▶ The **Lagrangian**

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i f_i(x) + \sum_j \mu_j h_j(x)$$

converts hard constraints into priced violations.

- ▶ The **dual function**

$$g(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu)$$

is always concave and always provides a lower bound on the primal optimum.

- ▶ **Weak duality** always holds:

$$d^* \leq p^*.$$

- ▶ **Strong duality** does not always hold, but for convex problems it is guaranteed by **Slater's condition**.

- ▶ Under strong duality, the **KKT conditions** characterize optimal primal-dual pairs.

- ▶ The optimal multiplier λ^* is also a **shadow price**: it measures the sensitivity of the optimal value to constraint relaxation.

Appendix: optimal control through the optimization lens

The KKT framework extends naturally from finite-dimensional optimization to optimization over **functions**.

Optimal control problem

Choose a control trajectory $u(\cdot)$ to minimize

$$J[u] = \int_0^T \ell(x(t), u(t)) dt + \Phi(x(T))$$

subject to the dynamics

$$\dot{x}(t) = f(x(t), u(t)), \quad x(0) = x_0.$$

Dictionary

- ▶ finite-dimensional variable $x \in \mathbb{R}^n \longleftrightarrow$ trajectory $u(\cdot)$ and state path $x(\cdot)$,
- ▶ equality constraint $h(x) = 0 \longleftrightarrow$ ODE constraint $\dot{x} - f(x, u) = 0$ for every time t ,
- ▶ multiplier vector $\mu \longleftrightarrow$ multiplier trajectory $p(t)$.

Finite-dimensional KKT vs. optimal control

Finite-dimensional optimization	Optimal control
Decision variable $x \in \mathbb{R}^n$	control $u(t)$ and state $x(t)$
Objective $f(x)$	$\int_0^T \ell(x, u) dt + \Phi(x(T))$
Constraint $h(x) = 0$	$\dot{x}(t) - f(x(t), u(t)) = 0$ for every t
Multiplier μ	costate trajectory $p(t)$
Stationarity $\nabla_x \mathcal{L} = 0$	first variation of the functional vanishes

Main message

Pontryagin's principle is the continuous-time analogue of the KKT conditions.

The functional Lagrangian

Write the dynamics as the constraint

$$f(x(t), u(t)) - \dot{x}(t) = 0 \quad \text{for all } t \in [0, T].$$

Assign a multiplier trajectory $p(t)$ to this continuum of equality constraints.

Lagrangian functional

$$\mathcal{L}[x, u, p] = \int_0^T \ell(x, u) dt + \Phi(x(T)) + \int_0^T p(t)^\top (f(x, u) - \dot{x}) dt.$$

Define the Hamiltonian

$$H(x, u, p) = \ell(x, u) + p^\top f(x, u).$$

Then

$$\mathcal{L}[x, u, p] = \int_0^T H(x, u, p) dt - \int_0^T p^\top \dot{x} dt + \Phi(x(T)).$$

What replaces the gradient in function space?

In finite dimensions, stationarity means

$$\nabla_x \mathcal{L}(x, \mu) = 0.$$

Why? Because for every direction v ,

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{L}(x + \varepsilon v, \mu) = 0.$$

In function space, we do the same thing:

- ▶ perturb $u(\cdot)$ to $u(\cdot) + \varepsilon \delta u(\cdot)$,
- ▶ perturb $x(\cdot)$ to $x(\cdot) + \varepsilon \delta x(\cdot)$,
- ▶ require the first variation to vanish for every perturbation.

Key idea

“gradient = 0” $\iff$ “first variation vanishes in every direction.”

Variation with respect to the control

Hold $x(\cdot)$ and $p(\cdot)$ fixed, and perturb

$$u(t) \mapsto u(t) + \varepsilon \delta u(t).$$

Only the Hamiltonian term depends on u , so

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{L}[x, u + \varepsilon \delta u, p] = \int_0^T (\nabla_u \ell(x, u) + J_u f(x, u)^\top p)^\top \delta u(t) dt.$$

But

$$\nabla_u \ell(x, u) + J_u f(x, u)^\top p = \nabla_u H(x, u, p).$$

So stationarity with respect to the control means

$$\int_0^T \nabla_u H(x, u, p)^\top \delta u(t) dt = 0 \quad \forall \delta u(\cdot).$$

Hence the integrand must vanish:

$$\boxed{\nabla_u H(x, u, p) = 0.}$$

More generally, if $u(t)$ is constrained to lie in a set $\mathcal{U}$, this becomes

$$u^*(t) \in \arg \min_{u \in \mathcal{U}} H(x^*(t), u, p^*(t)).$$

Variation with respect to the state: the integration-by-parts step

Now perturb the state:

$$x(t) \mapsto x(t) + \varepsilon \delta x(t), \quad \delta x(0) = 0$$

because the initial condition is fixed.

Differentiating the Lagrangian gives

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{L} = \int_0^T (\nabla_x \ell + J_x f^\top p)^\top \delta x \, dt - \int_0^T p^\top \dot{\delta x} \, dt + \nabla \Phi(x(T))^\top \delta x(T).$$

The problematic term is

$$- \int_0^T p^\top \dot{\delta x} \, dt.$$

Use integration by parts:

$$- \int_0^T p^\top \dot{\delta x} \, dt = - [p^\top \delta x]_0^T + \int_0^T \dot{p}^\top \delta x \, dt.$$

Since $\delta x(0) = 0$,

$$- [p^\top \delta x]_0^T = -p(T)^\top \delta x(T).$$

Costate equation and terminal condition

After integration by parts, the first variation becomes

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{L} = \int_0^T (\nabla_x \ell + J_x f^\top p + \dot{p})^\top \delta x \, dt + (\nabla \Phi(x(T)) - p(T))^\top \delta x(T).$$

For this to vanish for every perturbation $\delta x(\cdot)$, both pieces must vanish:

Costate equation

$$\dot{p}(t) = -\nabla_x \ell(x, u) - J_x f(x, u)^\top p(t) = -\nabla_x H(x, u, p).$$

Terminal condition

$$p(T) = \nabla \Phi(x(T)).$$

Together with the state equation

$$\dot{x}(t) = f(x(t), u(t)),$$

we obtain the continuous-time analogue of the KKT system.

Pontryagin's conditions

Let

$$H(x, u, p) = \ell(x, u) + p^\top f(x, u).$$

Then Pontryagin's maximum (or minimum) principle says that an optimal trajectory satisfies:

❶ **State equation**

$$\dot{x}^*(t) = f(x^*(t), u^*(t)), \quad x^*(0) = x_0.$$

❷ **Costate equation**

$$\dot{p}^*(t) = -\nabla_x H(x^*(t), u^*(t), p^*(t)), \quad p^*(T) = \nabla \Phi(x^*(T)).$$

❸ **Hamiltonian minimization**

$$u^*(t) \in \arg \min_{u \in \mathcal{U}} H(x^*(t), u, p^*(t)).$$

Analogy

These are exactly the KKT conditions in function space: primal feasibility, stationarity, and multiplier dynamics.

Pontryagin vs. HJB

There are two classical ways to characterize optimal control.

Pontryagin

Gives **necessary conditions**:

- ▶ state equation,
- ▶ costate equation,
- ▶ Hamiltonian minimization.

This is the continuous-time KKT viewpoint.

Hamilton-Jacobi-Bellman (HJB)

Works with the value function

$$V(x, t) = \inf_{u(\cdot)} \text{future cost.}$$

It satisfies the PDE

$$-\partial_t V(x, t) = \min_u \{ \ell(x, u) + \nabla_x V(x, t)^\top f(x, u) \}.$$

This is the dynamic-programming viewpoint.

Connection

Along an optimal trajectory,

$$p^*(t) = \nabla_x V(x^*(t), t).$$

So the costate is the continuous-time version of a shadow price.

Final takeaway

The unifying idea

Lagrange multipliers are not just algebraic tricks. They are:

- ▶ **certificates** of optimality,
- ▶ **supporting hyperplanes** in geometry,
- ▶ **shadow prices** in sensitivity analysis,
- ▶ and, in optimal control, **costate variables**.

What to remember

- 1 Build the Lagrangian.
- 2 Minimize over primal variables to get the dual function.
- 3 Maximize the dual function to get the best lower bound.
- 4 Under Slater, the lower bound is exact.
- 5 Under strong duality, KKT gives a complete optimality system.

Duality turns lower bounds into optimality certificates.